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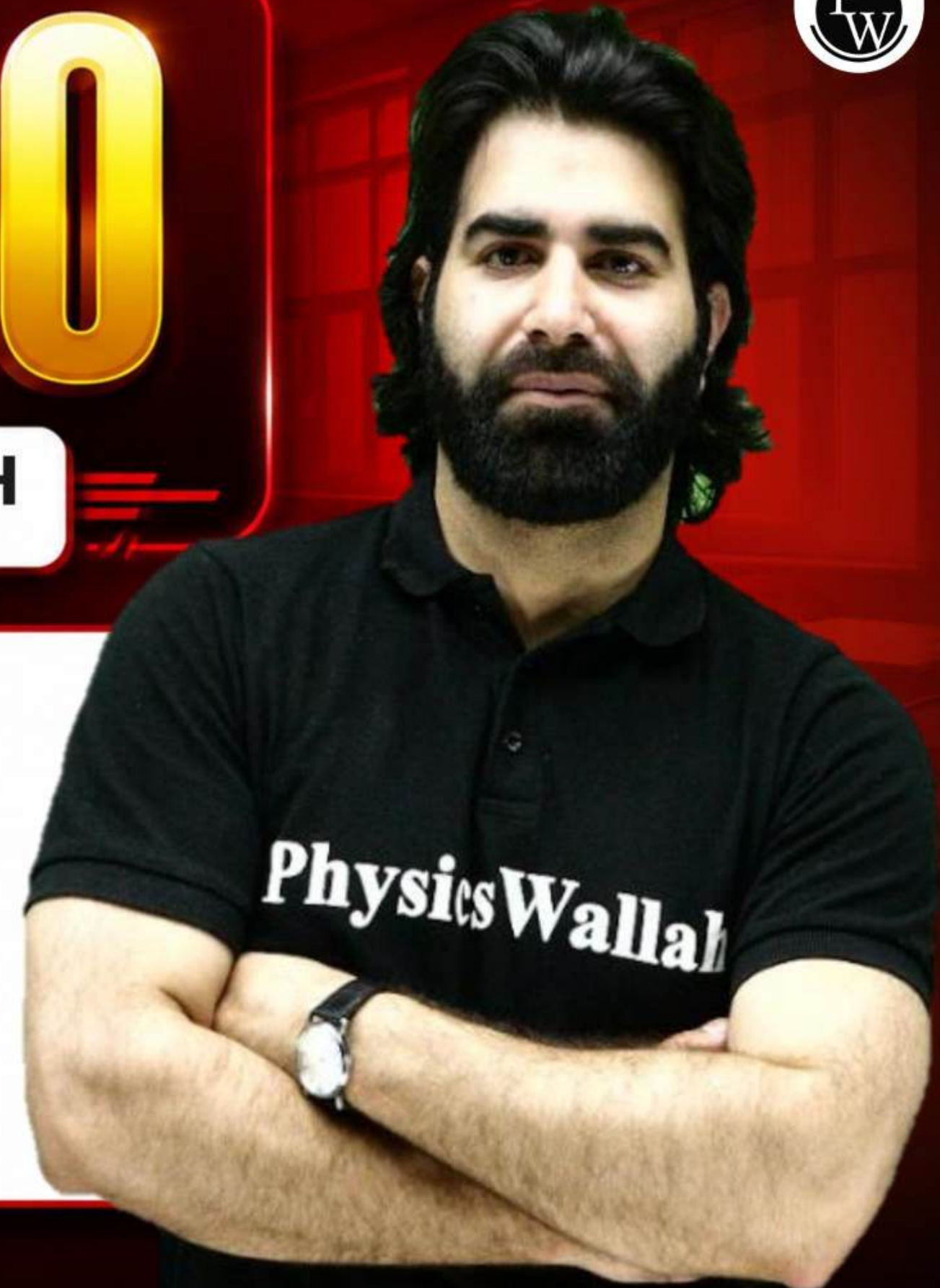
JKBOSE CLASS-11TH

✓ Cell The Unit Of Life

One Shot

Zoology

Adil Altaf Sir



Topics To Be Covered

- 1 Cell , Introduction, cell Theory.
- 2 Eukaryotic / Prokaryotic
- 3 → Cell Membrane, cell organelles.
- 4 → Nucleus.



CELL THE UNIT OF LIFE

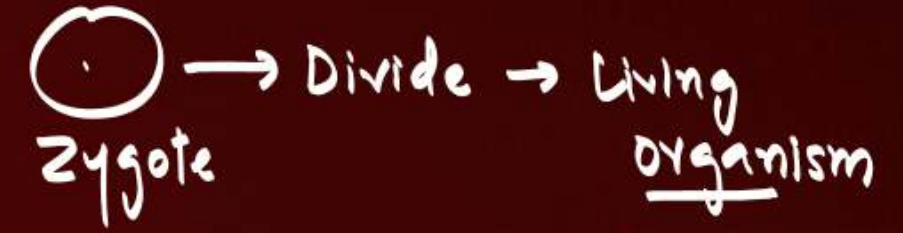
$$\text{Life} = n \times \text{cell} \Rightarrow \text{Life} = 1 \times \text{cell}$$

$n = 1, 2, 3 \dots$



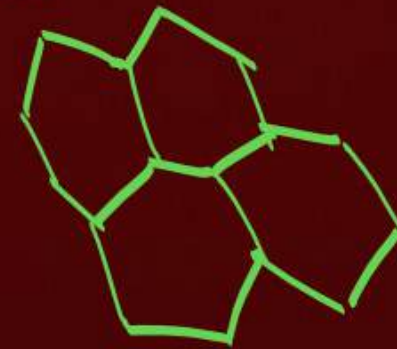
- ✓ A cell is the fundamental, structural and functional unit of all living organisms. - بنیاد -

- ✓ Robert Hooke: Discovered cell. (1665) → observed cork: (Part of Bark) - Slice (thin)



- ✓ Anton Von Leeuwenhoek: First observed and described a live cell.

- The invention of the compound & electron microscopes revealed all the structural details of the cell.



→ Actually saw, cell walls



CELL THEORY

Ql. Imp. Botanist.



- ❑ **Matthias Schleiden (1838)** observed that all plants are composed of different kinds of cells. *→ Zoologist*
- ❑ **Theodore Schwann (1839)** found that cells have a thin outer layer (plasma membrane). He also found that plant cells have cell wall. He proposed a hypothesis that animals and plants are composed of cells and products of cells.
- ❑ Schleiden & Schwann formulated the cell theory.
- ❑ **Rudolf Virchow (1855)** first explained that cells divide and new cells are formed from pre-existing cells (Omnis cellula-e cellula). He modified the cell theory. *→ cells from pre existing cell.*
- ❑ Cell theory states that:



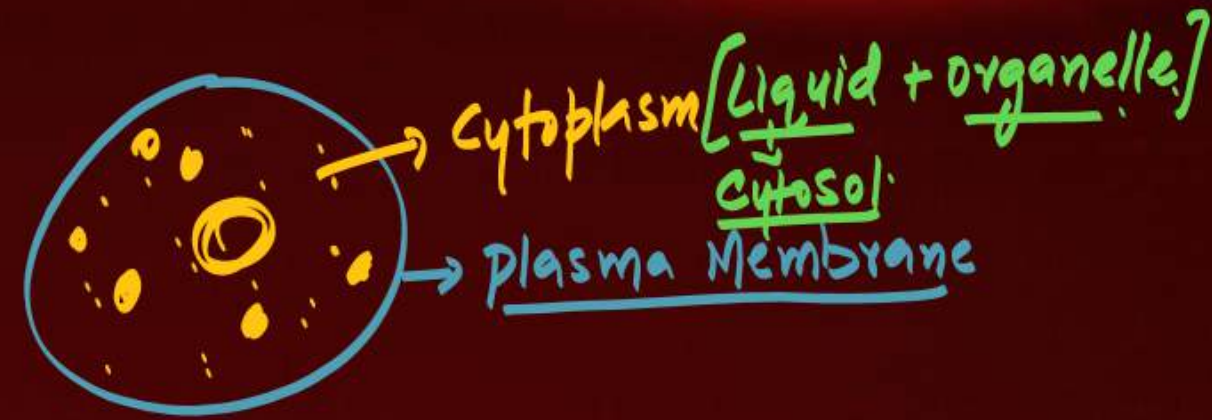
CELL THEORY



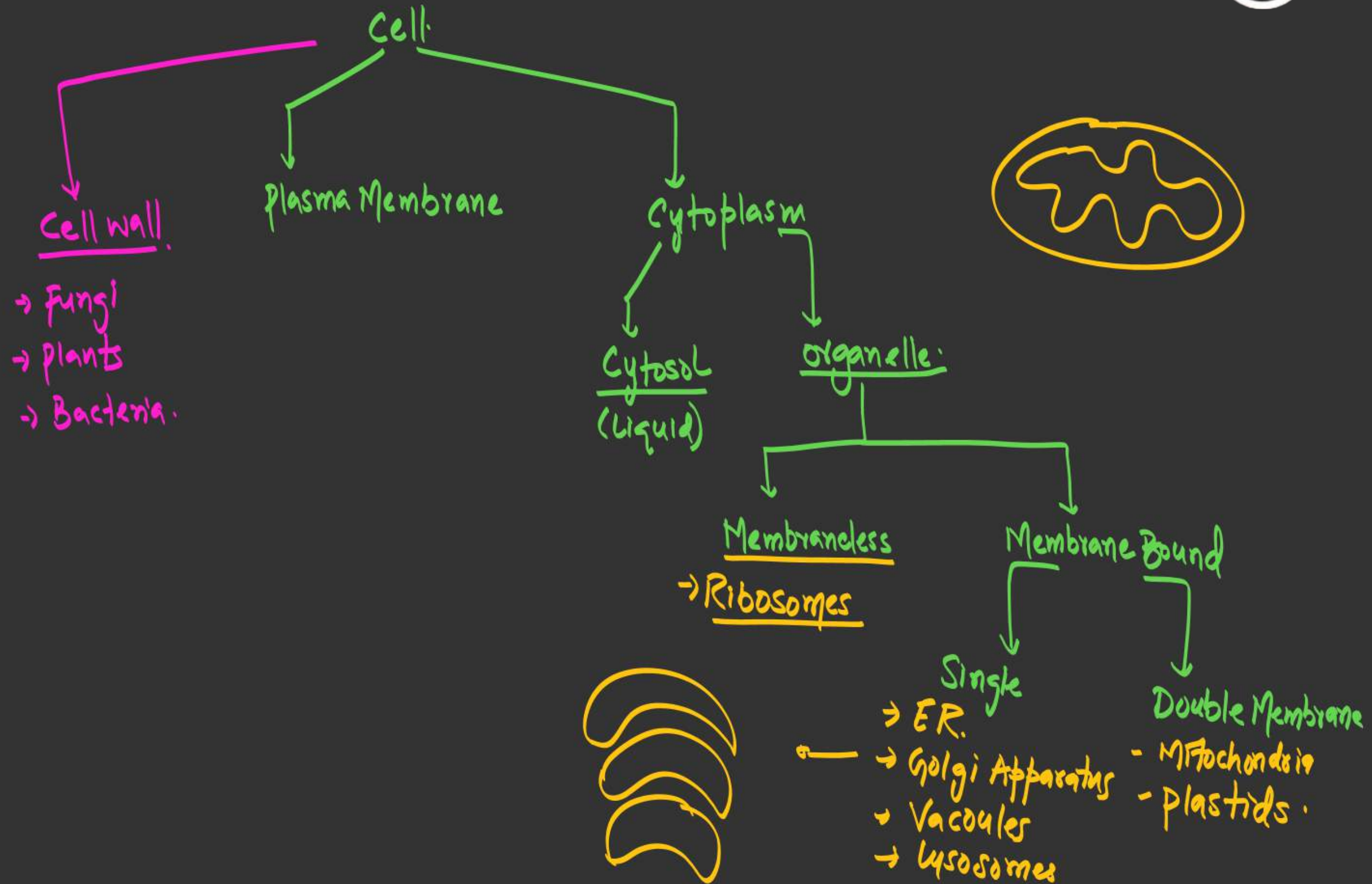
- (i) ✓ All living organisms are composed of cells and products of cells.
- (ii) ✓ All cells arise from pre-existing cells.



AN OVERVIEW OF CELL



- All cells contain
- Cytoplasm: A semi-fluid matrix where cellular activities and chemical reactions occur. This keeps the cell in 'living state'.
- Ribosomes: Non-membrane bound organelles seen in cytoplasm, chloroplasts, mitochondria & on rough ER.
- Cells differ in size, shape and activities.
- Smallest cells: Mycoplasmas ($0.3\mu\text{m}$ in length). → Belong to Moneran (Bacteria)
→ without cell wall.
- Largest isolated single cell: Egg of ostrich.
- Longest cells: E.g. Nerve cell.
- Size of bacteria: 3 to $5\mu\text{m}$ (Typical: 1 to $2\mu\text{m}$).
- Human RBCs are about $7.0\mu\text{m}$ in diameter.





AN OVERVIEW OF CELL



- ❑ Based on the functions, shape of cells may be disc-like, polygonal, columnar, cuboid, thread like, or irregular.
- ❑ Cells are 2 types: Prokaryotic cells & Eukaryotic cells.
 - On the Basis of Nucleus.



PROKARYOTIC CELLS

→ No nucleus / Nuclear Membrane

- They have no membrane bound nucleus and organelles.
- They include bacteria, blue-green algae, mycoplasma & PPLO (Pleuro Pneumonia Like Organisms).
 - Cyanobacteria
 - photosynthetic
- They are generally smaller and multiply more rapidly than the eukaryotic cells.
- They vary in shape & size. E.g. Bacteria have 4 basic shapes: Bacillus, Coccus, Vibrio and Spirillum.

↓
Round

↓
Comma,

↓
(Spiral)

↓
Rod like



PROKARYOTIC CELLS



Cell organelles in prokaryotic cells

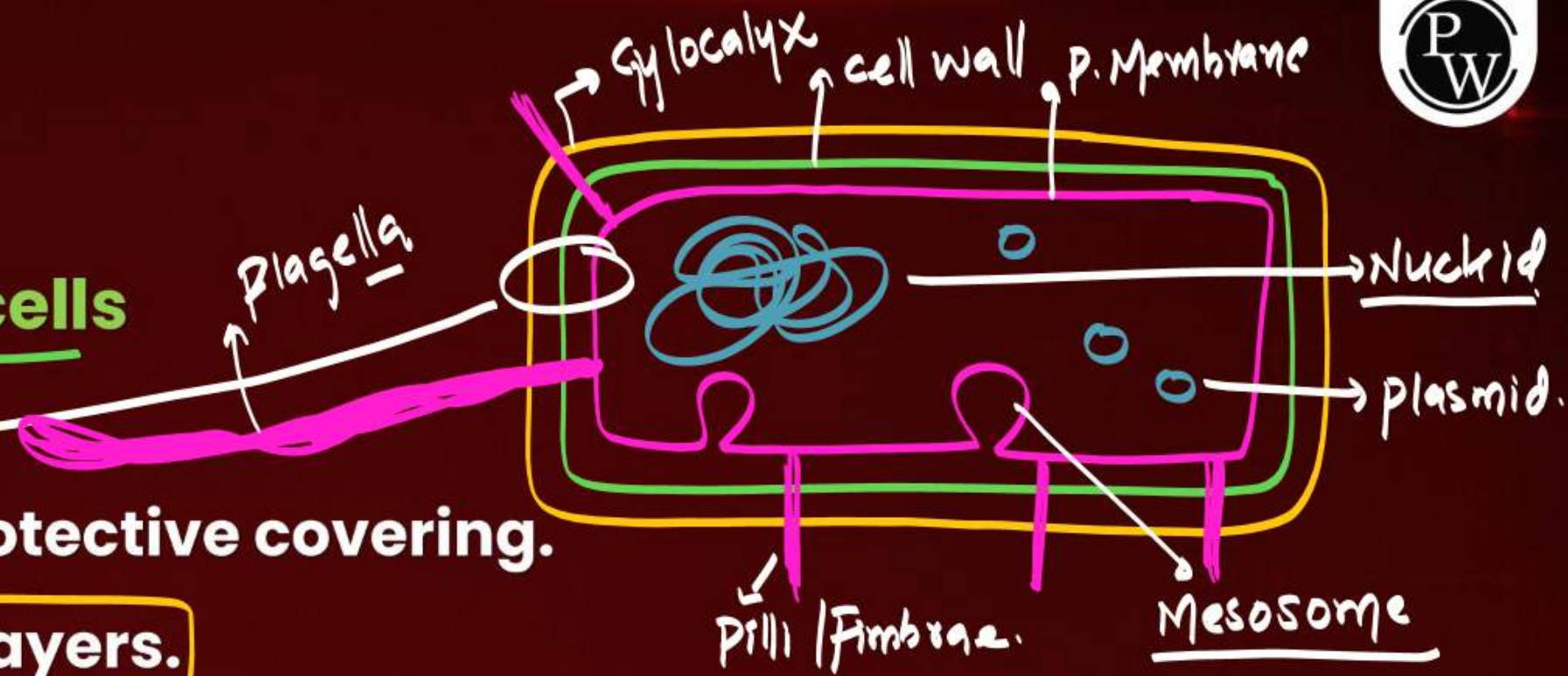
Cell Envelope

✓ It is a chemically complex protective covering.

It is made of 3 tightly bound layers.

① Glycocalyx: Outer layer. Its composition and thickness vary in different bacteria. It may be a slime layer (loose sheath) or capsule (thick & tough).

② Cell wall: Middle layer. Seen in all prokaryotes except mycoplasma. It gives shape to the cell and provides a structural support to prevent the bract erbium from bursting or collapsing.





PROKARYOTIC CELLS



○ ③ Plasma membrane: Inner layer. It is semi-permeable in nature and interacts with the outside. This is structurally similar to that of the eukaryotes.

Selectively permeable

□ ✓ Based on the types of the cell envelopes and response to Gram staining (developed by Gram), bacteria are 2 types:

○ Gram positive: They take up and retain the gram stain.

○ Gram negative: They do not retain the gram stain

□ → Mesosomes & Chromatophores (Membranous structures)

□ Mesosome is formed by the infoldings of plasma membrane. It includes vesicles, tubules & lamellae.





PROKARYOTIC CELLS



Mitochondria → Mesosome

Chloroplast → chromatophores

- ❑ **Functions: Mesosomes help**
 - In cell wall formation. ✓
 - In DNA (chromosome) replication. ✓
 - ❑ In distribution of chromosomes to daughter cells. ✓
 - **In respiration and secretion processes.**
 - To increase the surface area of the plasma membrane and enzymatic content.
- ❑ **Chromatophores are pigment-containing membranous infoldings in some prokaryotes (e.g. cyanobacteria).**



PROKARYOTIC CELLS



1. Nucleoid

- ❑ It is formed of non-membranous (naked) circular genomic DNA (single chromosome/ Genetic material) & protein.
- ❑ Many bacteria have small circular DNA (plasmid) outside the genomic DNA. It gives some unique phenotypic characters (e.g. resistance to antibiotics) to bacteria.

1. Flagella

- ❑ These are thin ^(Thread) filamentous extensions from the cell wall of motile bacteria. Their number and arrangement are varied in different bacteria.



PROKARYOTIC CELLS



- ❑ Bacterial flagellum has 3 parts – filament, hook and basal body. The filament is the longest portion and extends from the cell surface to the outside.

1. Pili and Fimbriae

- ❑ These are surface structures that have no role in motility.
- ❑ Pili (sing. Pilus) are elongated tubular structures made of a special protein (pilin).
- ❑ Fimbriae are small bristle like fibres sprouting out of the cell. In some bacteria, they help to attach the bacteria to rocks in streams and to the host tissues.



PROKARYOTIC CELLS



Ribosomes

→ Protein factories

- ❑ They are associated with plasma membrane of prokaryotes.
- ❑ They are about 15 nm by 20 nm in size.
- ❑ They are made of 2 subunits – **50S & 30S** (Svedberg's unit). They together form 70S prokaryotic ribosomes.

→ Prokaryotic 70S - Subunits $\left\{ \begin{array}{l} \rightarrow 50S \\ \rightarrow 30S \end{array} \right.$
↓
S = Svedberg Unit.
= Coefficient of Sedimentation



PROKARYOTIC CELLS



(S = sedimentation coefficient; a measure of density & size).

Function: Ribosomes are the site of translation (**protein synthesis**). Several ribosomes may attach to a single mRNA to form a chain called polyribosomes (polysome). Ribosomes translate the mRNA into proteins.



PROKARYOTIC CELLS



1. Inclusion Bodies

- ❑ These are non-membranous, stored reserve material seen freely in the cytoplasm of prokaryotic cells.
- ❑ E.g. phosphate granules, cyanophycean granules and glycogen granules, gas vacuoles etc.
- ❑ Gas vacuoles are found in blue green and purple and green photosynthetic bacteria.



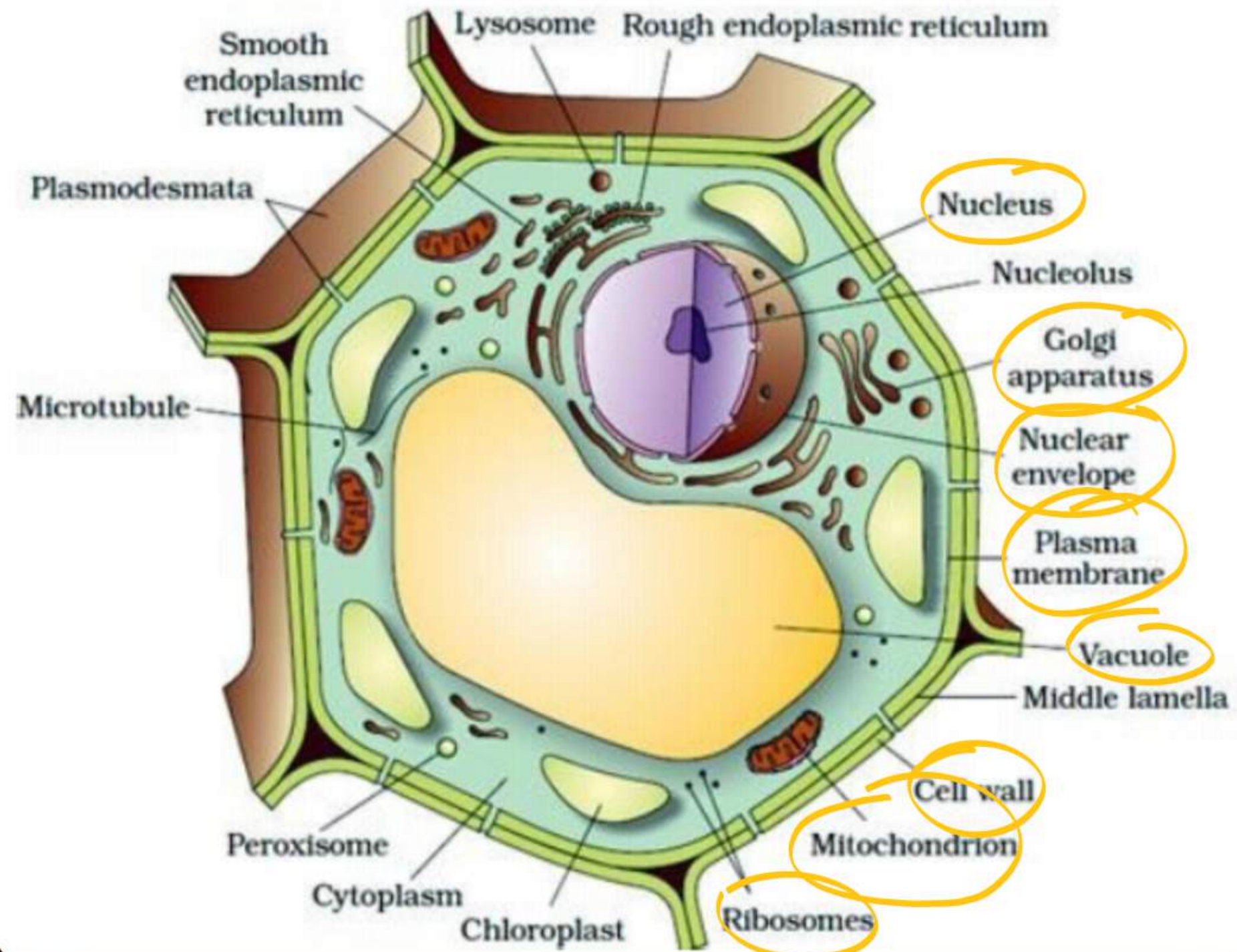
EUKARYOTIC CELLS



- ❑ They have well organized membrane bound nucleus and organelles.
- ❑ Presence of membranes gives clear compartmentalization of cytoplasm.
- ❑ Their genetic material is organized into chromosomes. *→ Humans = 46*
- ❑ They have complex locomotory & cytoskeletal structures.

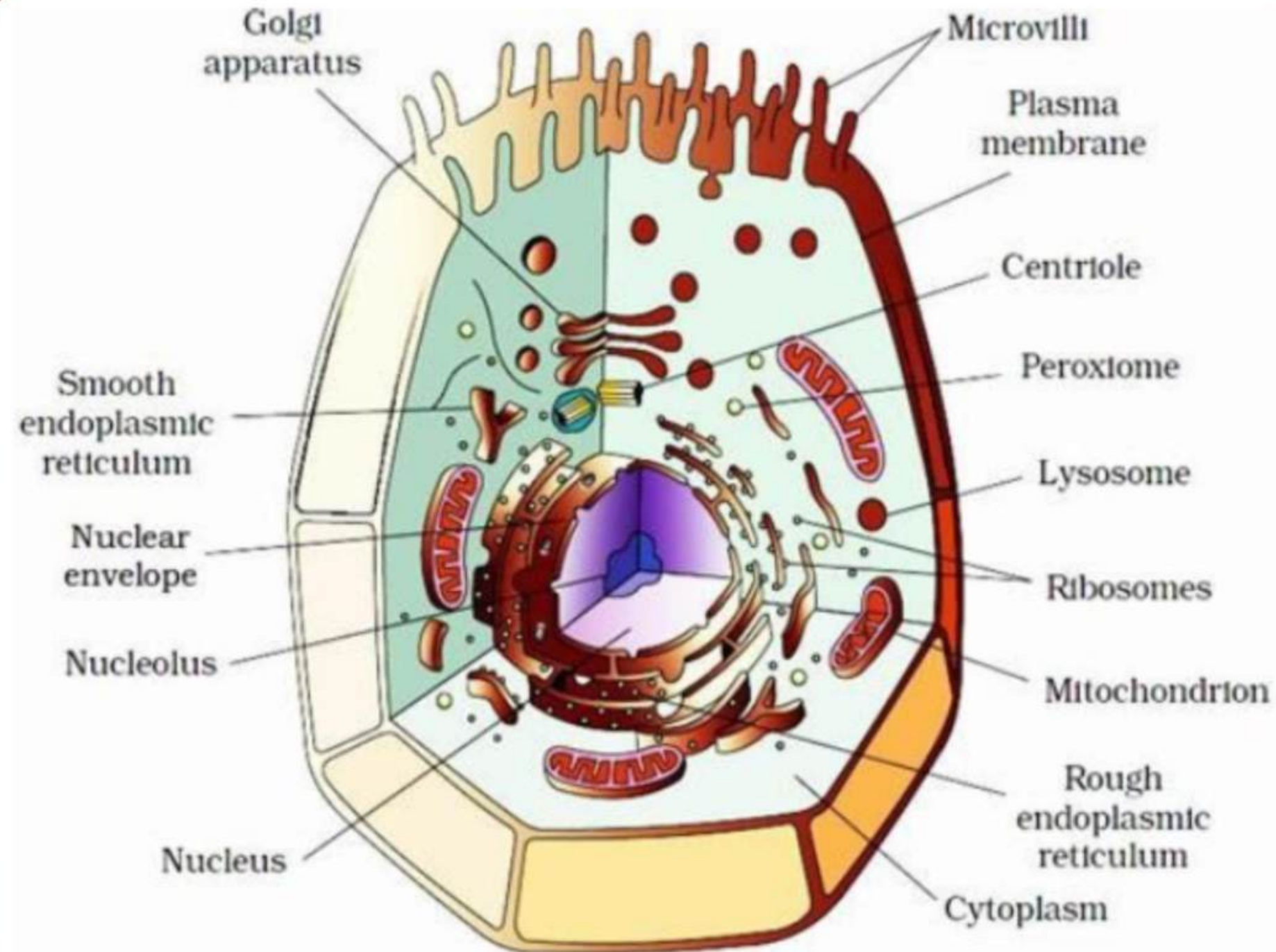


Plant cell and Animal cell





Plant cell and Animal cell



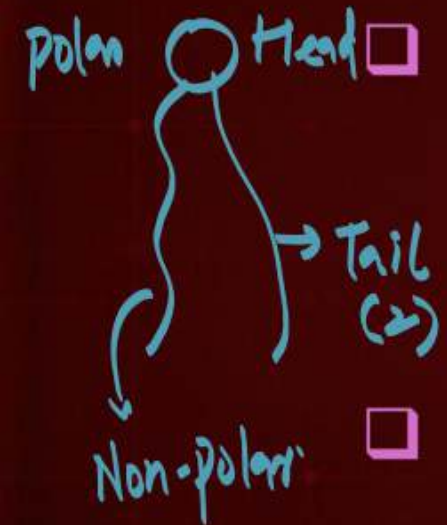


Cell organelles in eukaryotic cells



1. Cell Membrane (Plasma Membrane)

- Chemical studies on human RBCs show that cell membrane is composed of a lipid bilayer, protein & carbohydrate.
- Lipids (mainly phosphoglycerates) have outer polar head and the inner hydrophobic tails. So the non-polar tail of saturated hydrocarbons is protected from the aqueous environment.
- Ratio of protein and lipid varies in different cells. E.g. In human RBC, membrane has 52% protein and 40% lipids.
- Based on the ease of extraction, membrane proteins are 2 types:
- Integral proteins:** Partially or totally buried in membrane.





Cell organelles in eukaryotic cells

- ❑ **Peripheral proteins:** Lie on the surface of membrane.
- ❑ **Fluid mosaic model of cell membrane:** Proposed by Singer & Nicolson (1972). According to this, the quasifield nature of lipid enables lateral movement of proteins within the overall bilayer. This ability to move within the membrane is measured as its fluidity.

Forp

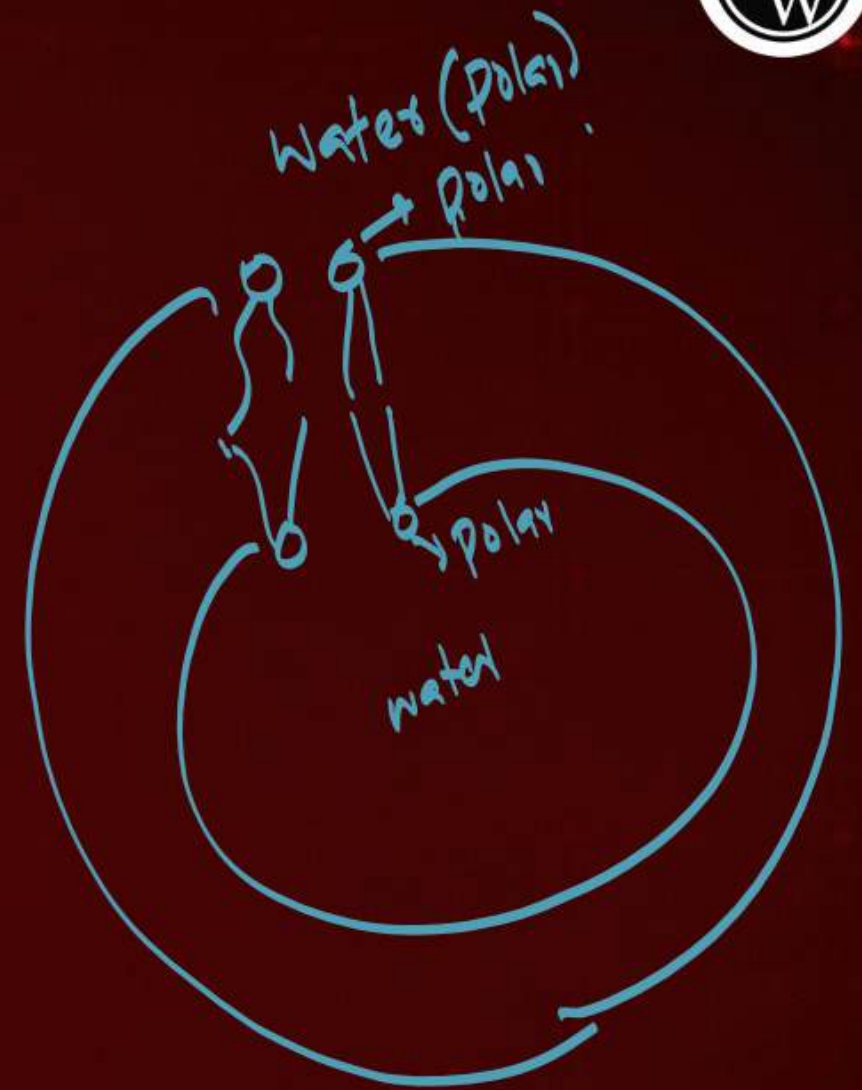
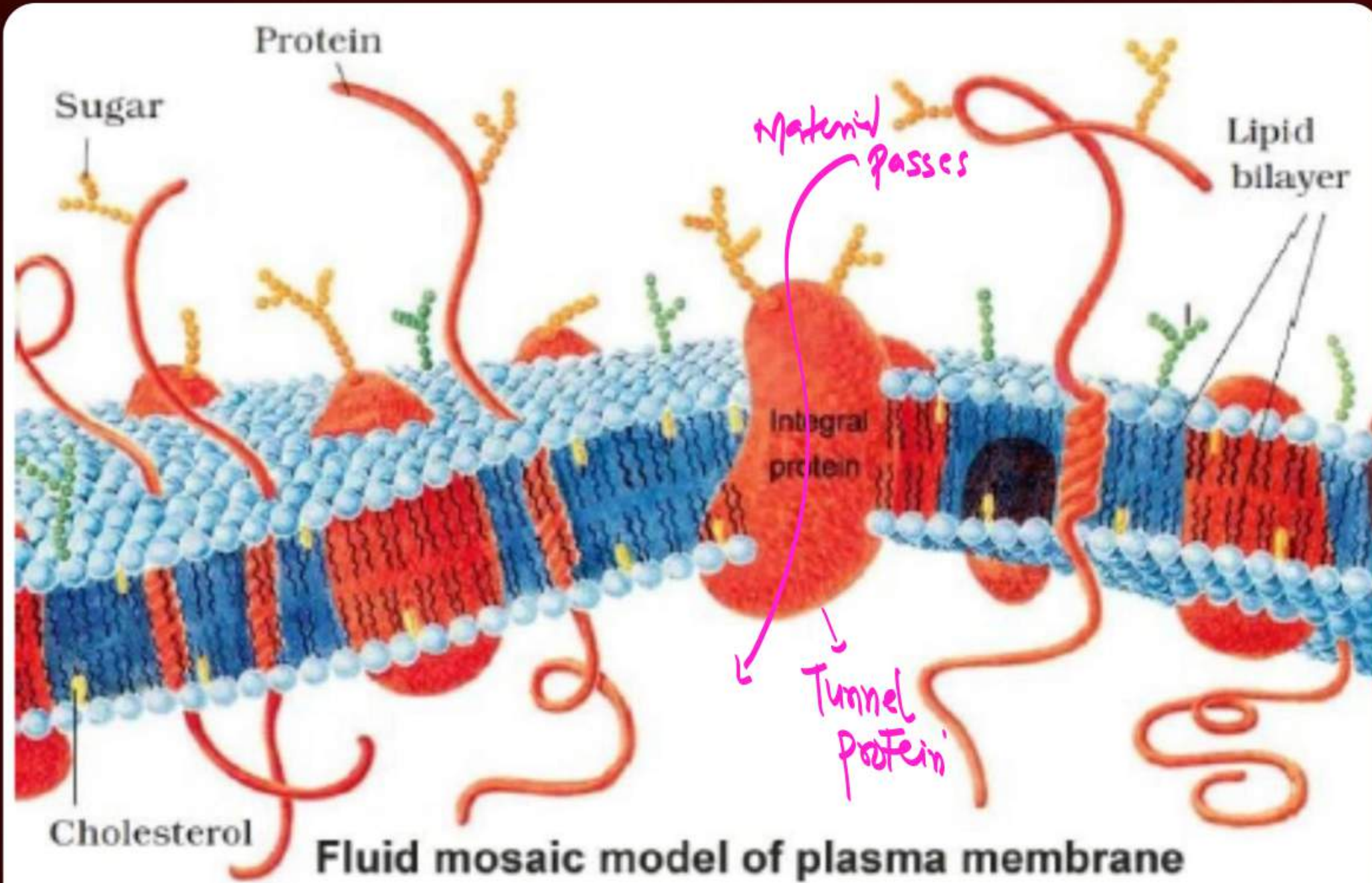
→ P.M = Phospholipid Bilayer
= Proteins at places

→ Proteins → Integral
→ Peripheral

→ According to them → (Proteins are like Icebergs in the sea of lipid)



Cell organelles in eukaryotic cells





Cell organelles in eukaryotic cells

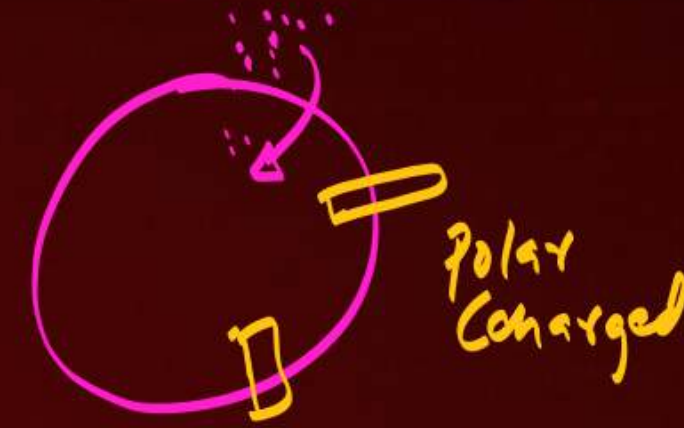
Functions:

- ❑ Transport of the molecules. The membrane is selectively permeable to some molecules present on either side of it.
- ❑ Due to the fluid nature, the plasma membrane can help in cell growth, formation of intercellular junctions, secretion, endocytosis, cell division etc.





Cell organelles in eukaryotic cells



Types of Transport

1. **Passive transport:** It is the movement of molecules across the membrane along the concentration gradient (i.e., from higher concentration to the lower) without the expenditure of energy. It is 2 types:
 - a. **Simple diffusion:** It is the movement of neutral solutes across the membrane.
 - b. **Osmosis:** It is the movement of water by diffusion across the membrane.
- ❑ **Polar molecules** cannot pass through the non-polar lipid bilayer. So they require membrane carrier protein for transport.

No Energy
ATP



Cell organelles in eukaryotic cells

2. **Active transport:** It is the movement of molecules across the membrane against the concentration gradient (i.e. from lower to the higher concentration) with the expenditure of energy (ATP is utilized). E.g. Na^+/K^+ pump.

2. Cell Wall

- ☐ ✓ It is a non-living rigid structure found outer to the plasma membrane of fungi and plants. + Bacteria.

- ☐ Cell wall of Algae is made of cellulose, galactans, mannans and minerals like CaCO_3 . In other plants, it consists of cellulose, hemicellulose, pectins and proteins. (Higher)

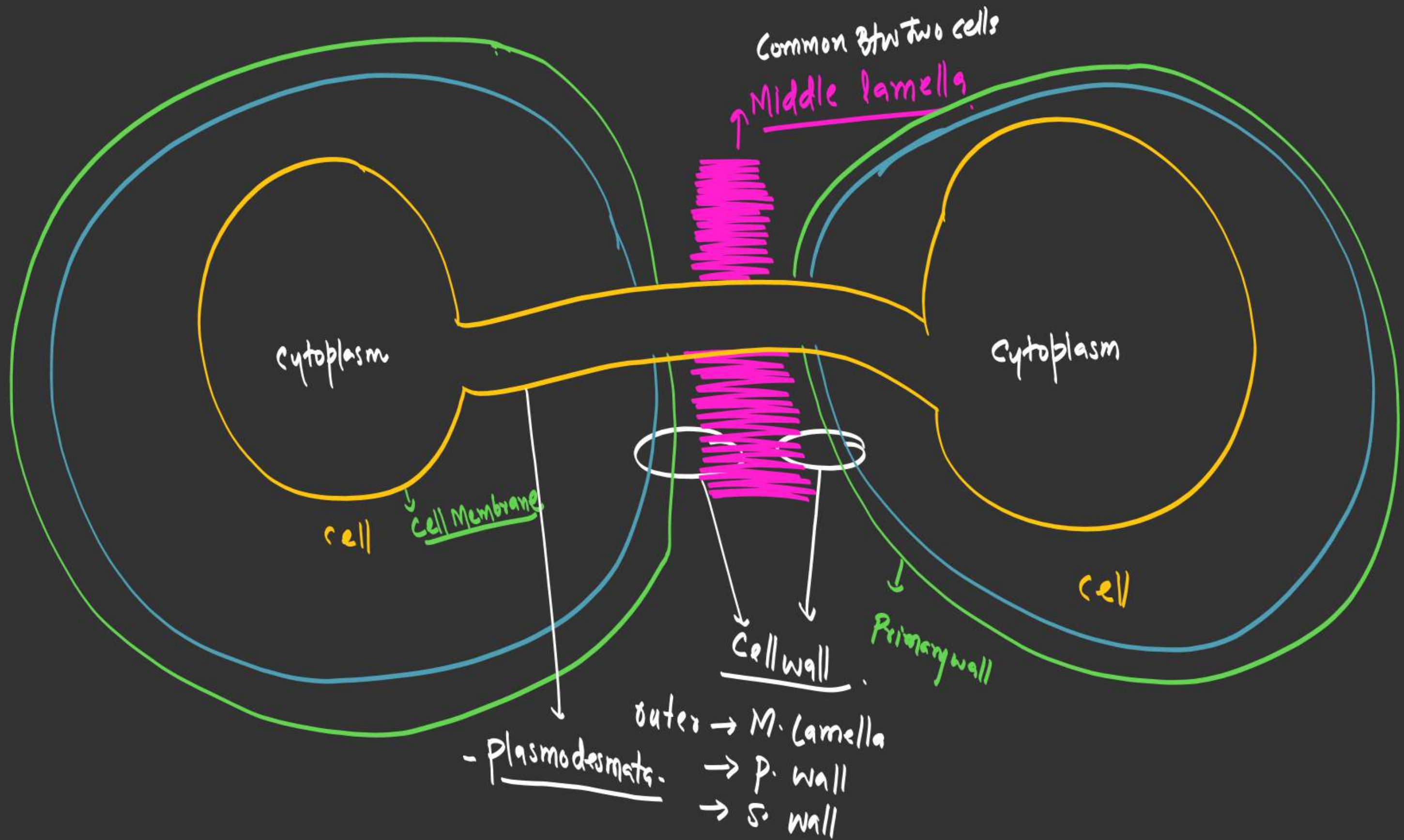
Composition

C H P P
↓ ↓ ↓ ↓



Cell organelles in eukaryotic cells

- ❑ Cell wall of a young plant cell (primary wall) is capable of growth. It gradually diminishes as the cell matures and the secondary wall is formed on the inner side (towards membrane).
- ❑ The middle lamella is a layer containing calcium pectate which glues the neighboring cells together. Cell wall and middle lamellae may be traversed by plasmodesmata. It connects the cytoplasm of neighbouring cells.



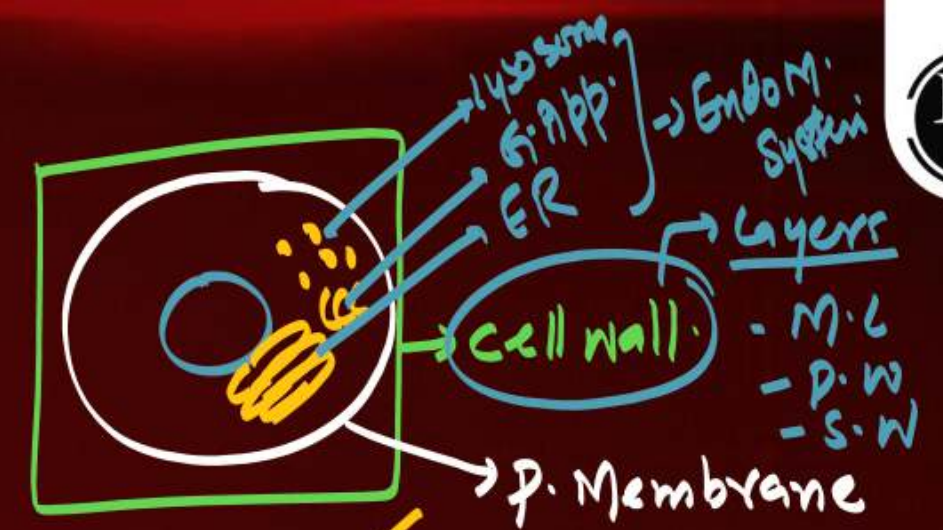


Cell organelles in eukaryotic cells



Functions:

- It gives shape to the cell. ✓
- It protects the cell from mechanical damage & infection.
- It helps in cell-to-cell interaction. ✓
- It acts as barrier to undesirable macromolecules. ✓



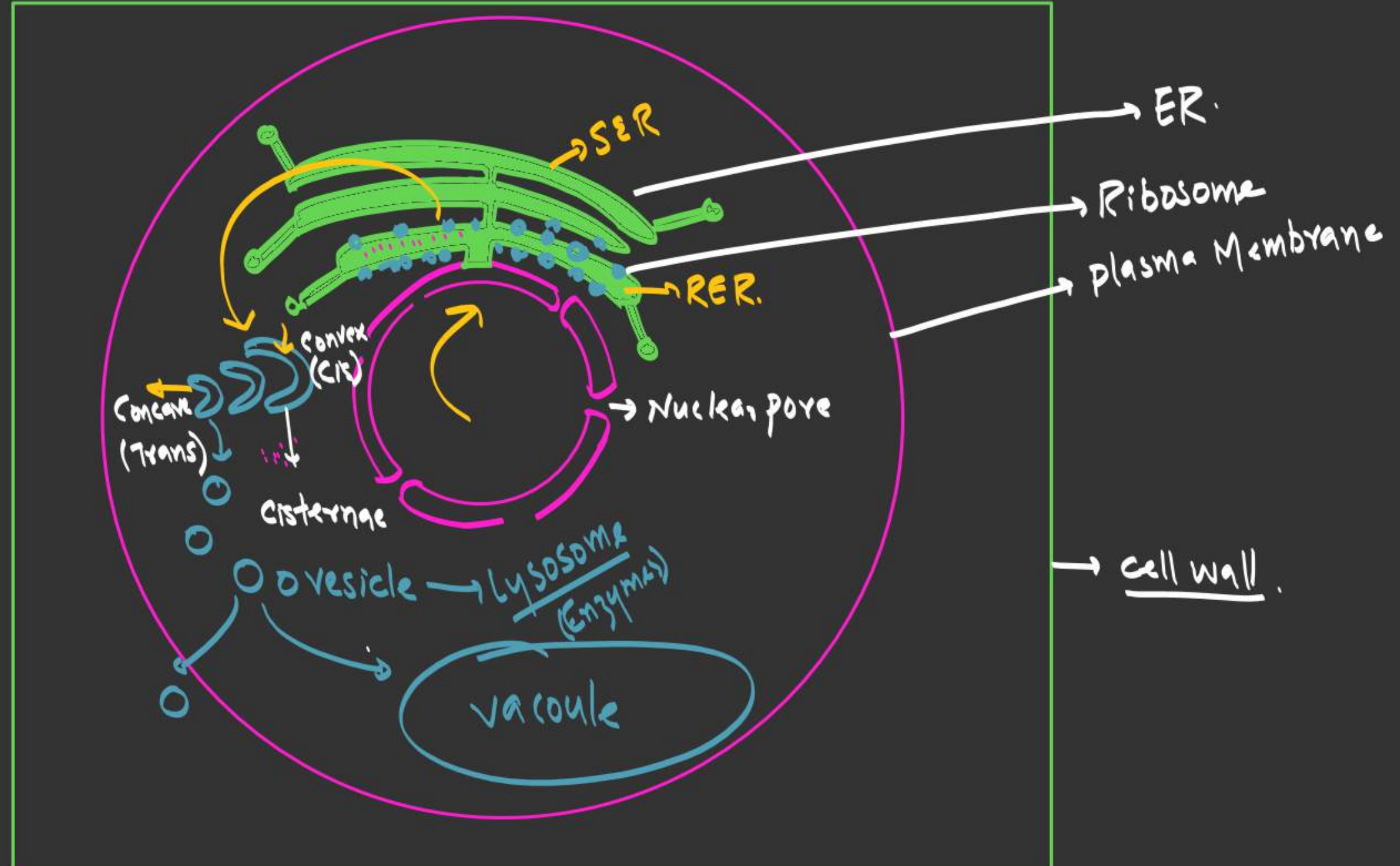
3. Endomembrane System

- It is a group of membranous organelles having coordinated functions.
- They include endoplasmic reticulum (ER), Golgi complex, lysosomes and vacuoles.



Endoplasmic Reticulum (ER) ✓

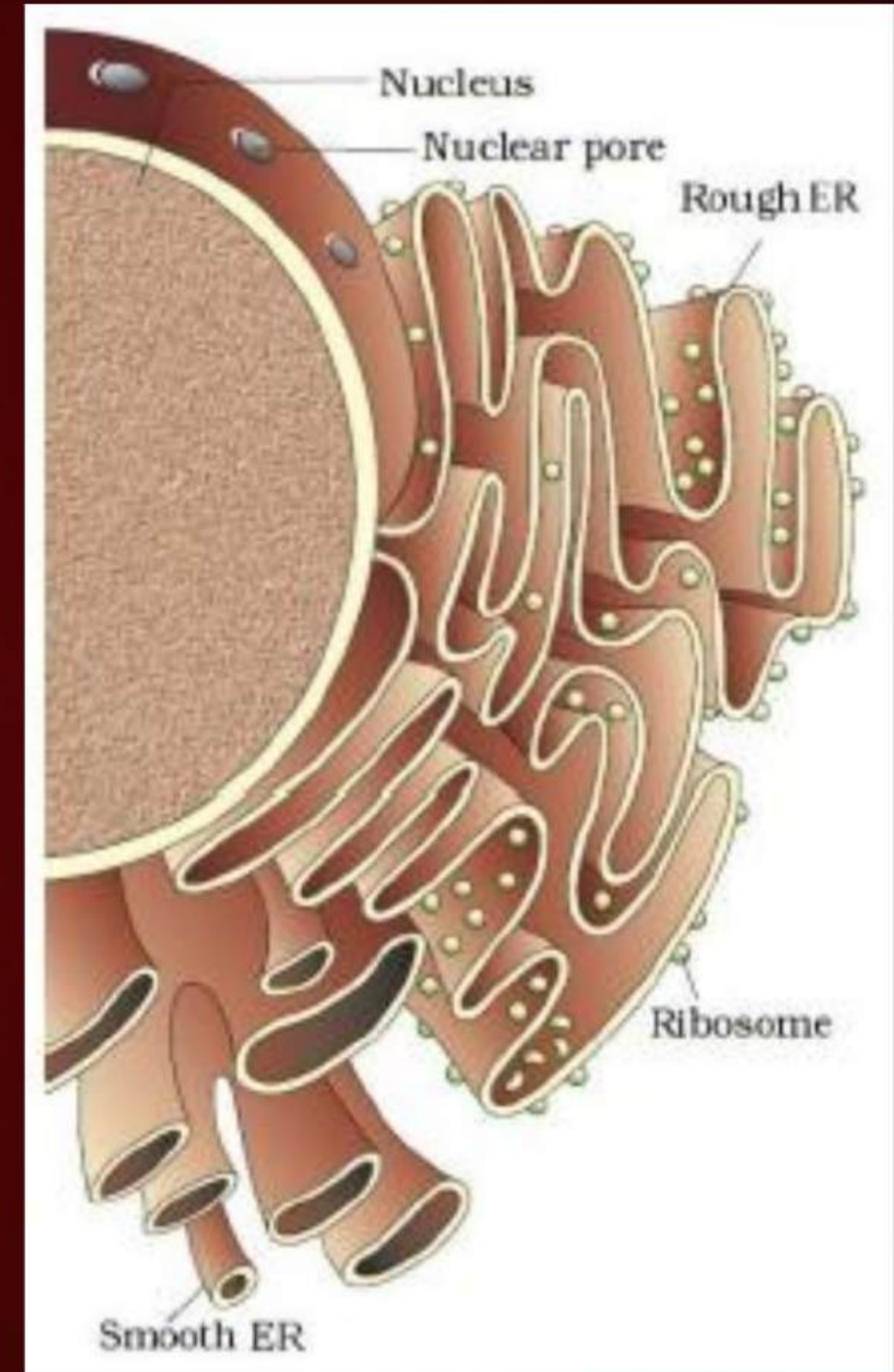
- ❑ These are a network of tiny tubular structures scattered in the cytoplasm.
- ❑ ER divides the intracellular space into 2 compartments: luminal (inside ER) & extra luminal (cytoplasm).
- ❑ Endoplasmic reticulum is 2 types:
 - a. **Rough endoplasmic reticulum (RER):**
 - ❑ Bear ribosomes on their surface. RER is frequently observed in the cells actively
 - ❑ involved in protein synthesis and secretion. They extend to the outer membrane of the nucleus.





Endoplasmic Reticulum (ER)

- b. Smooth endoplasmic reticulum (SER): Ribosomes are absent. SER is the major site for synthesis of lipid. In animal cells lipid-like steroidal hormones are synthesized in SER.





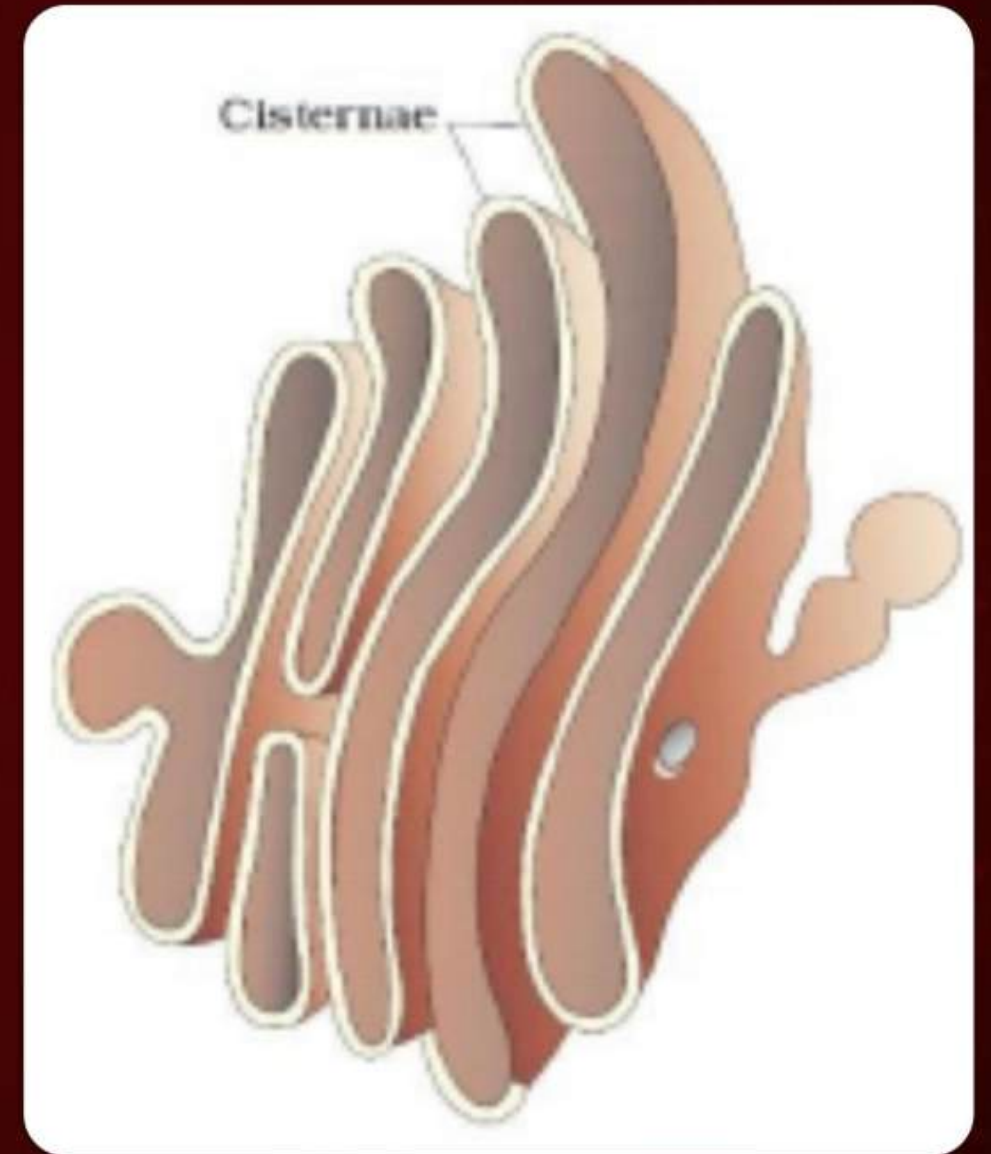
Golgi Apparatus



- ❑ **Densely stained reticular structures near the nucleus.**
- ❑ **First observed by Camillo Golgi (1898).**
- ❑ **They consist of flat, disc-shaped sacs (cisternae) of $0.5\text{--}1.0\mu\text{m}$ diameter. These are stacked parallelly.**
- ❑ **Cisternae are concentrically**
- ❑ **arranged with convex cis (forming) face and concave trans (maturing) face. Cis & trans faces are totally different, but interconnected.**

Function of Golgi apparatus:

- ❑ **Secretes materials to intra-cellular targets or outside the cell.**





Golgi apparatus



Hydrolysis
↓
H₂O Breakdown

Materials to be packaged as vesicles from the ER fuse with the cis face and move towards the trans face. This is why Golgi apparatus remains in close association with the endoplasmic reticulum.

- ❑ Proteins synthesized by ribosomes on the ER are modified in the cisternae of Golgi apparatus before they are released from its trans face.

- ❑ Formation of glycoproteins and glycolipids.

Lysosomes



- ❑ These are membrane bound vesicular structures formed by the process of packaging in the Golgi apparatus. ✓
- ❑ Lysosomal vesicles contain almost all types of hydrolytic enzymes (hydrolases—lipases, proteases, carbohydrase's). They are active at acidic pH. They digest carbohydrates, proteins, lipids and nucleic acids.



Vacuoles

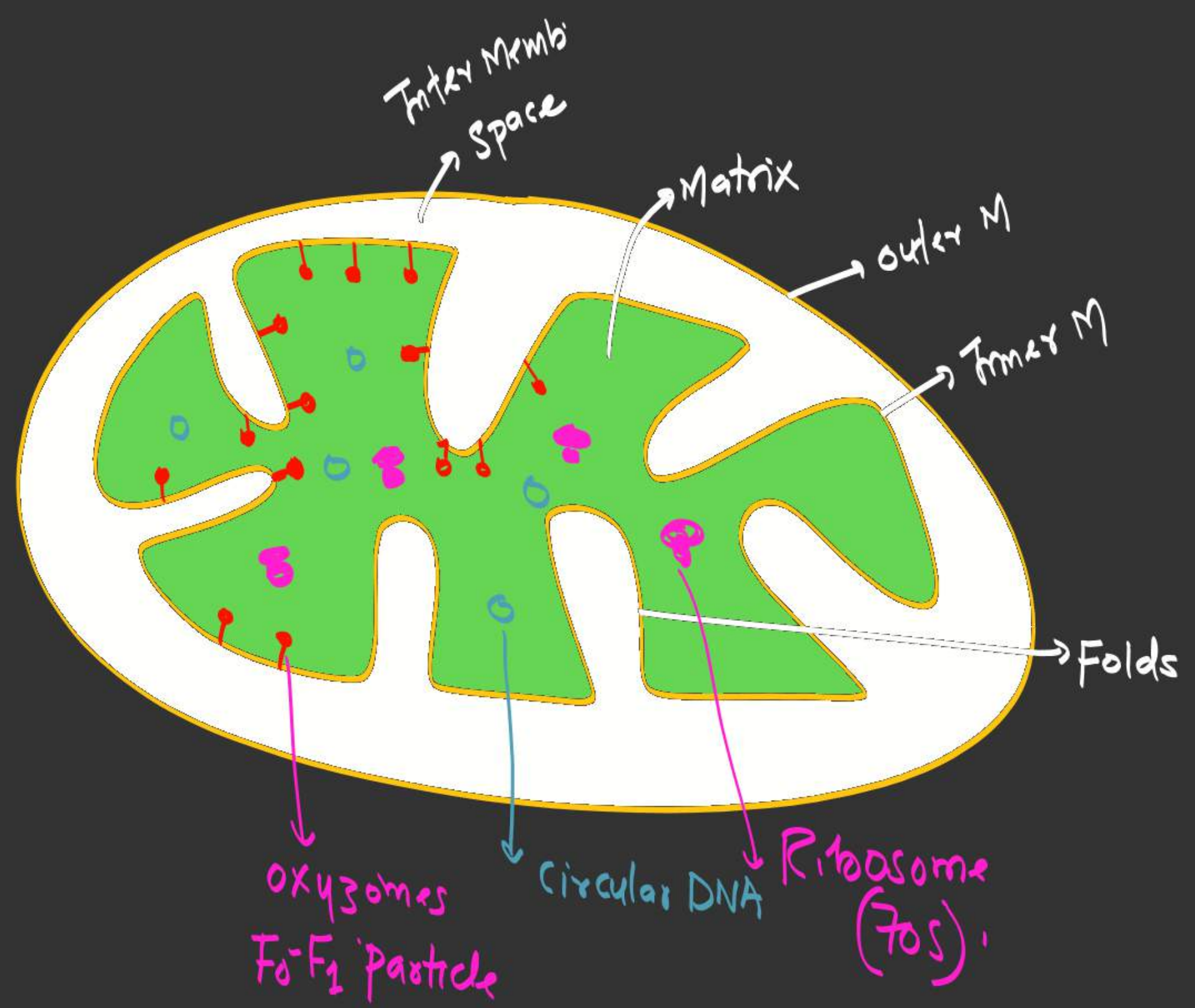
- ❑ **These are the membrane-bound space found in the cytoplasm. It contains water, sap, excretory product and other materials not useful for the cell.**
- ❑ **Vacuole is bound by a single membrane called tonoplast.**
- ❑ **In plant cells, the vacuoles can occupy up to 90% of the volume of the cell.**
- ❑ **In plants, the tonoplast facilitates the transport of ions and other materials against concentration gradients into the vacuole. Hence their concentration is higher in the vacuole than in the cytoplasm.**
- ❑ **In Amoeba, the contractile vacuole helps for excretion.**
- ❑ **In many cells (e.g. protists), food vacuoles are formed by engulfing the food particles.**

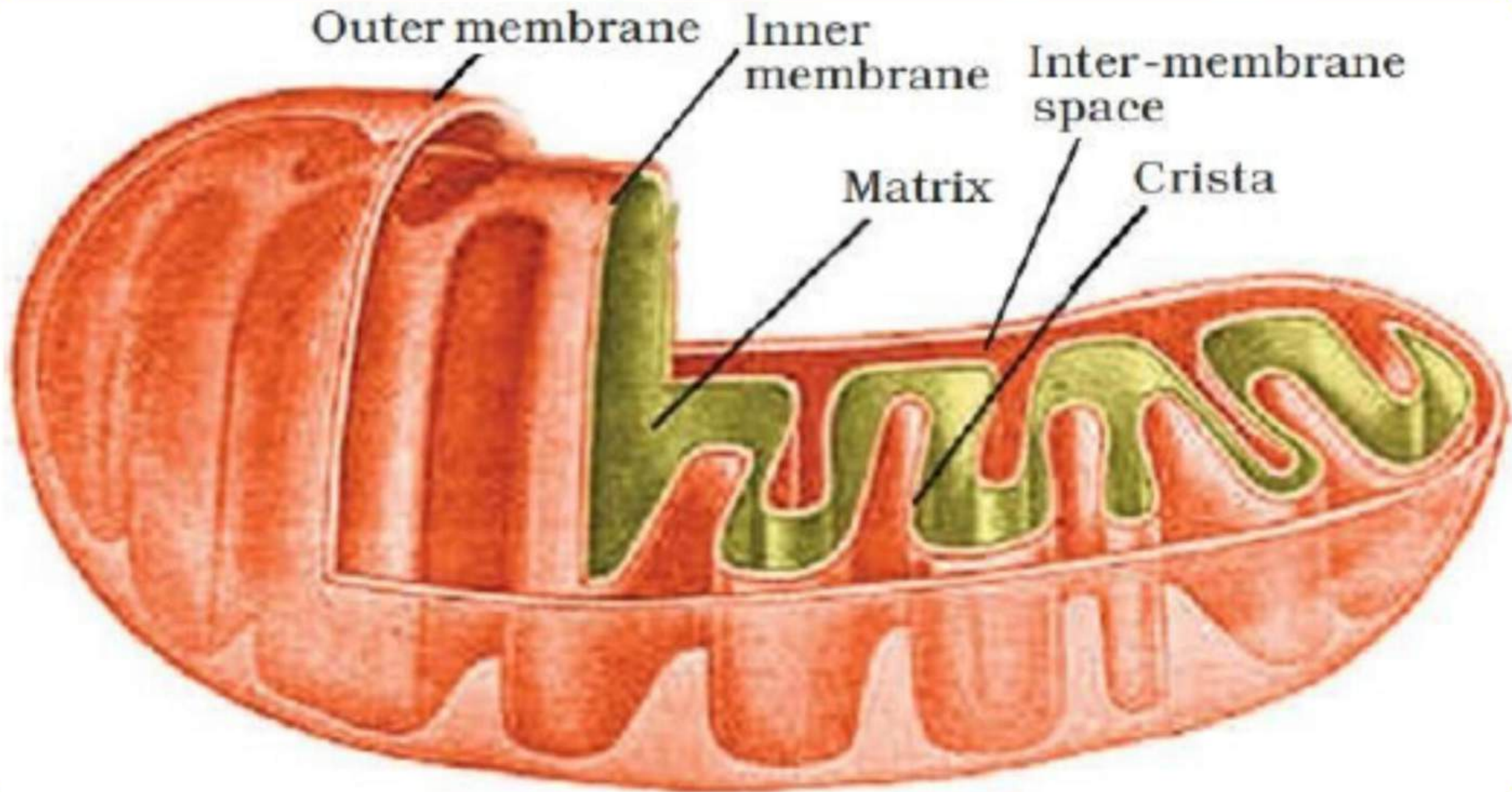


4. Mitochondria

→ powerhouse of cell
ATP
Aerobic Respiration

- ❑ Mitochondria are clearly visible only when stained.
- ❑ Number, shape and size of mitochondria per cell are variable depending on the physiological activity.
- ❑ It is sausage-shaped or cylindrical having a diameter of $0.2-1.0\mu\text{ m}$ (average $0.5\mu\text{ m}$) and length $1.0-4.1\mu\text{ m}$.







- ❑ A mitochondrion is a double membrane-bound structure with the outer membrane and the inner membrane. It divides lumen into 2 aqueous compartments, i.e., the outer compartment and the inner compartment (**matrix**).
- ❑ Inner membrane forms a number of infoldings (**cristae**) towards the matrix. They increase the surface area.
- ❑ ✓ The two membranes have their own specific enzymes associated with the mitochondrial function.
- ❑ ✓ Matrix possesses a circular DNA, a few RNA molecules, ribosomes (70S) and components for protein synthesis.
- ❑ The mitochondria divide by fission.



- ❑ **Function:** Mitochondria are the sites of aerobic respiration. They produce energy in the form of ATP. So they are called 'power houses' of the cell.

5. Plastids

- ❑ **Plastids are found in all plant cells and in euglenoides.**
- ❑ **Large sized. Easily observable under the microscope.**
- ❑ **They contain some pigments.**
- ❑ **Based on the type of pigments, plastids are 3 types:**

green a. Chloroplasts: Contain chlorophyll and carotenoid pigments. They trap light energy for photosynthesis.

Coloured b. Chromoplasts: Contain fat soluble carotenoid pigments like carotene, xanthophylls etc. This gives a yellow, orange or red colour.

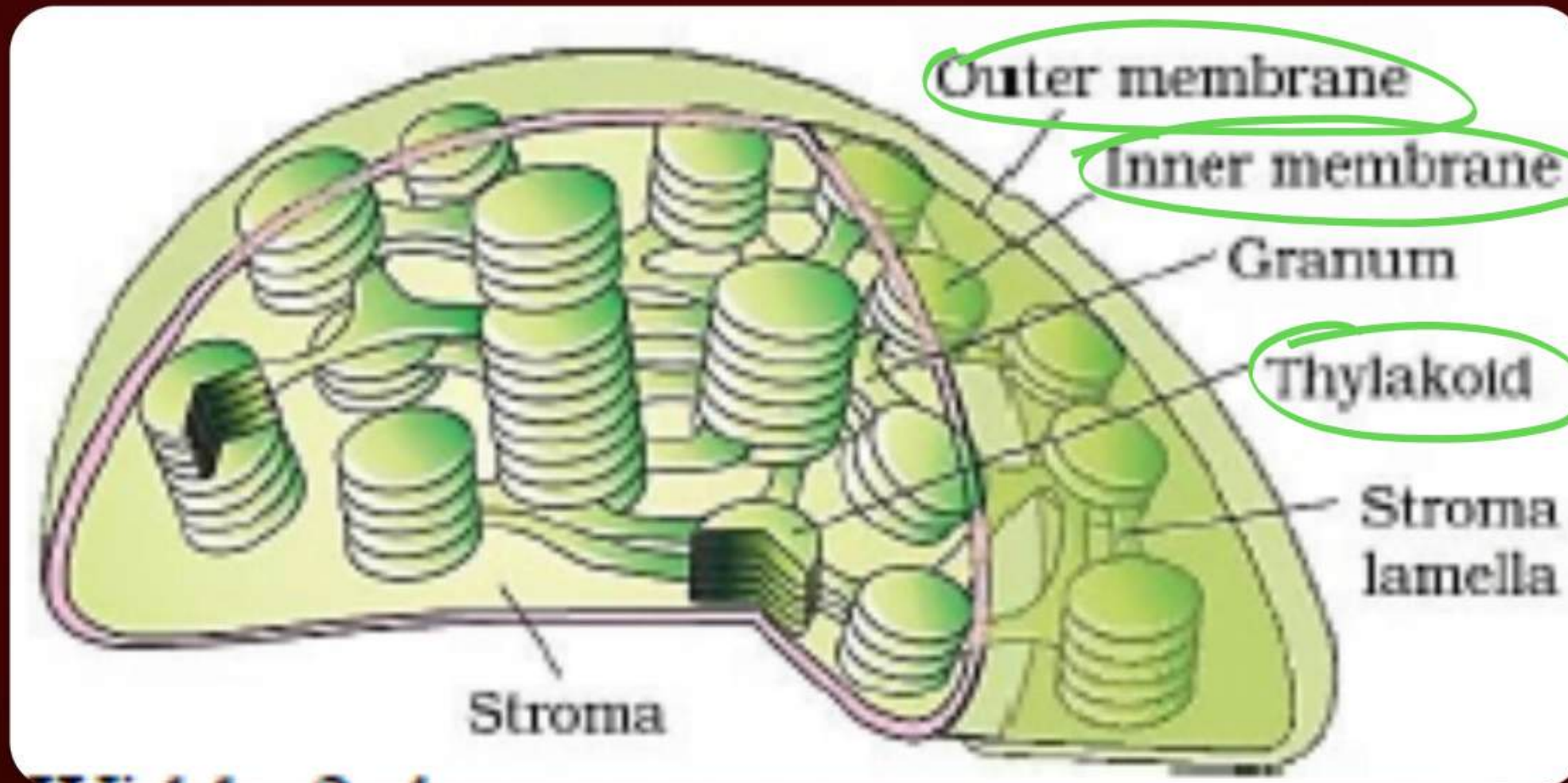


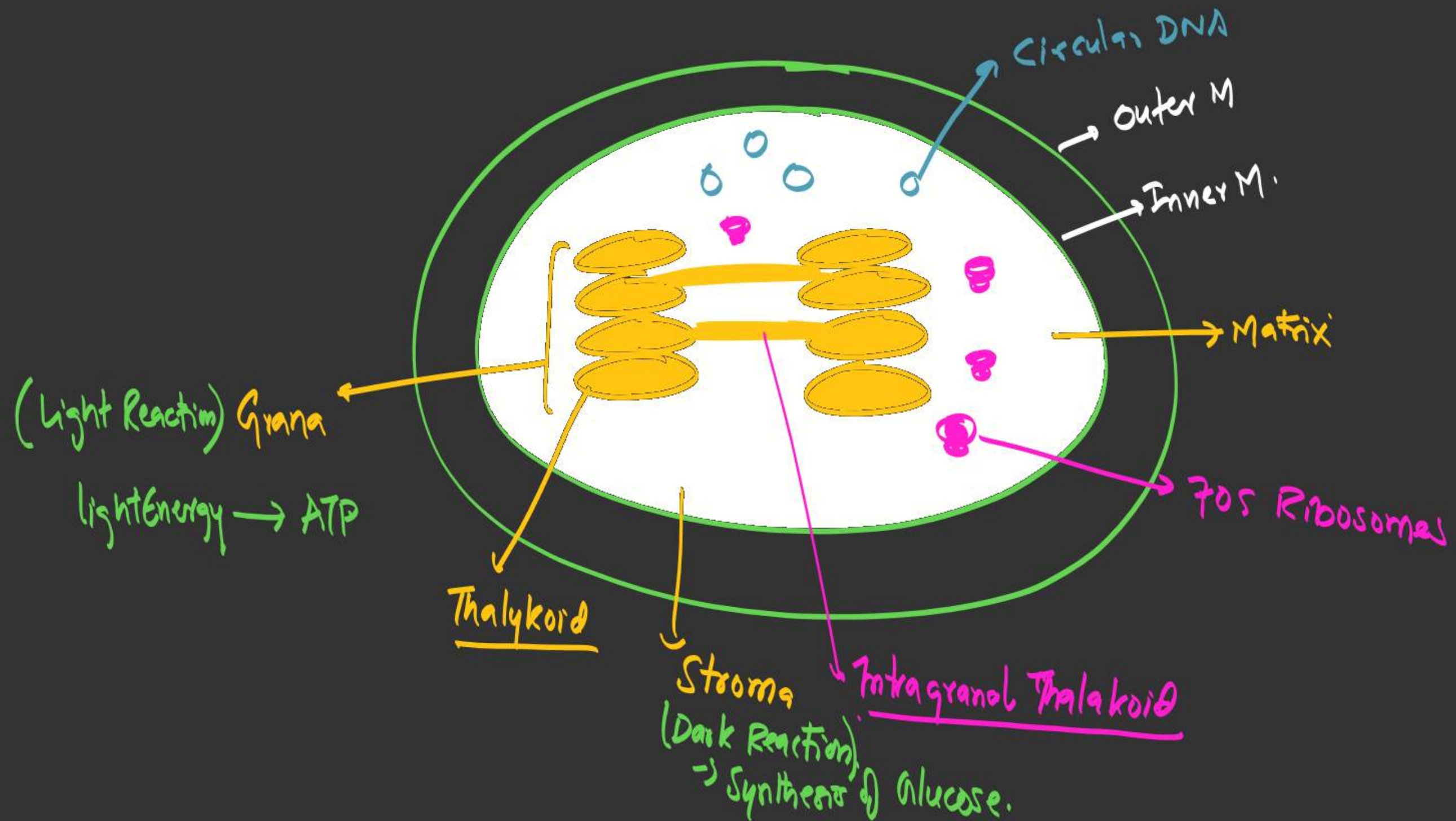
- c. **Leucoplasts:** These are colourless plastids of varied shapes and sizes with stored nutrients. They include:
- ❑ **Amyloplasts:** Store starch. E.g. potato.
 - ❑ **Elaioplasts:** Store oils and fats.
 - ❑ **Aleuroplasts:** Store proteins.



Chloroplasts:

- These are double membrane bound organelles mainly found in the mesophyll cells of the leaves. *Green parts*
- These are lens shaped, oval, spherical, discoid or ribbon-like organelles







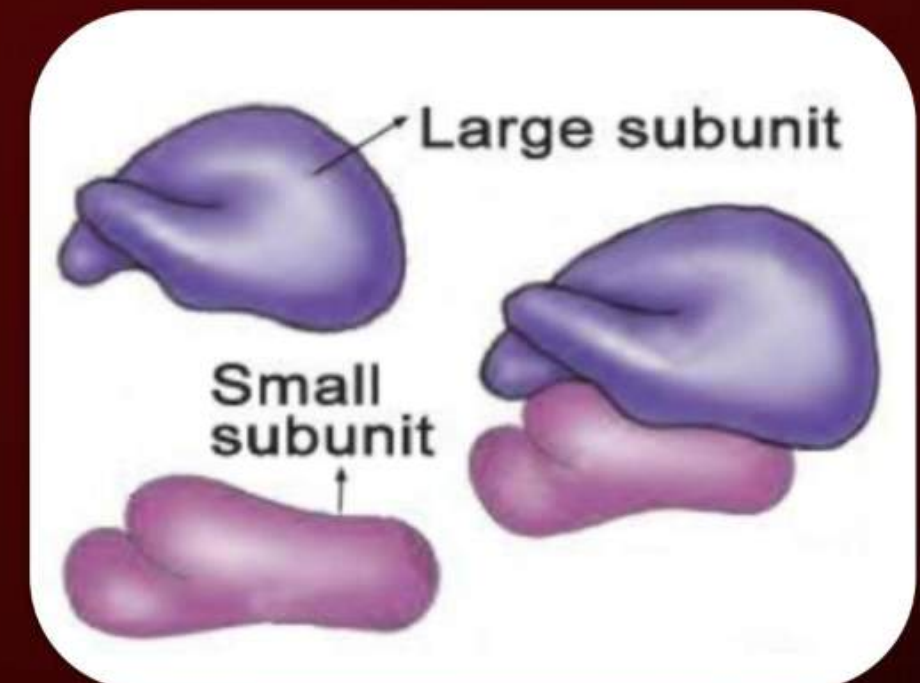
Chloroplasts:

- ❑ **Length: $5-10\mu\text{ m}$. Width: $2-4\mu\text{ m}$.**
- ❑ **Their number varies from 1 (e.g. Chlamydomonas) to 2040 per cell in the mesophyll.**
- ❑ **Inner membrane of chloroplast is less permeable.**
- ❑ **The space limited by the inner membrane of the chloroplast is called stroma. It contains many organized flattened membranous sacs called thylakoids.**
- ❑ **Membrane of thylakoids encloses a space called lumen.**
- ❑ **Chlorophyll pigments are present in the thylakoids.**
- ❑ **Thylakoids are arranged in stacks called grana or the intergranal thylakoids.**



Chloroplasts:

- ❑ There are flat membranous tubules called the stroma lamellae connecting the thylakoids of the different grana.
- ❑ The stroma contains small, double-stranded circular DNA molecules, ribosomes and enzymes for the synthesis of carbohydrates and proteins.
- ❑ The ribosomes of the chloroplasts are smaller (70S) than the cytoplasmic ribosomes (80S).





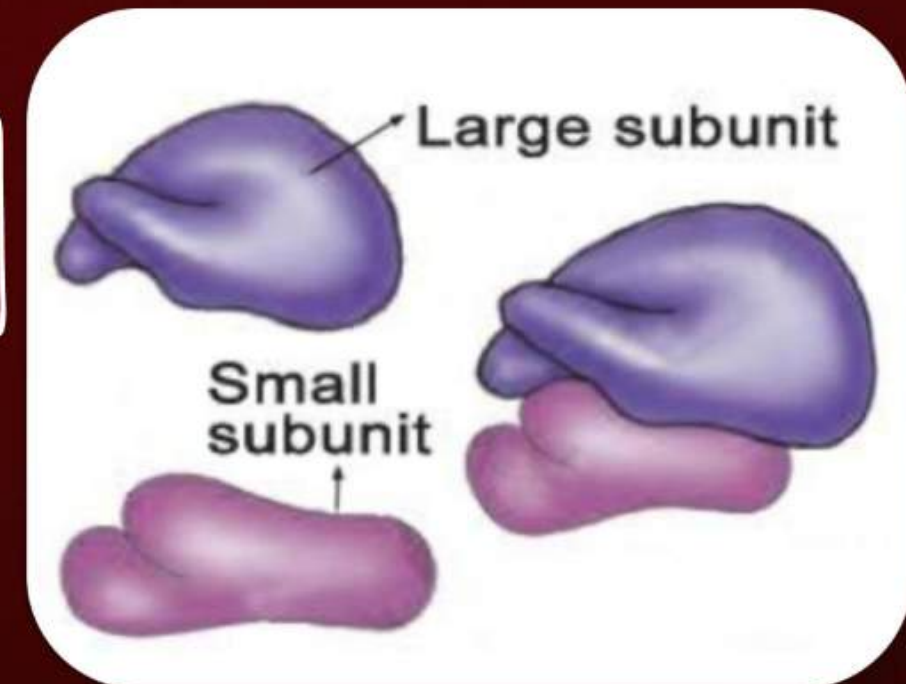
Chloroplasts:



6. Ribosomes

- They are non-membranous granular structures composed of ribonucleic acid (RNA) & proteins.
- It is first observed by George Palade (1953).
- Eukaryotic ribosome has 2 subunits– 60S (large subunit) and 40S (small subunit). They together form 80S.

→ Ribosomes $\begin{matrix} \text{70S} \\ \text{80S} \end{matrix} \xrightarrow{\text{Subunits}} \begin{matrix} 50\text{S} + 30\text{S} \\ 60\text{S} + 40\text{S} \end{matrix} \begin{matrix} \text{[Prokaryotic} \\ \text{mitochondria} \\ \text{chloroplasts}] \\ \text{[Eukaryotic} \\ \text{cytoplasm]} \end{matrix}$





7.

Cytoskeleton

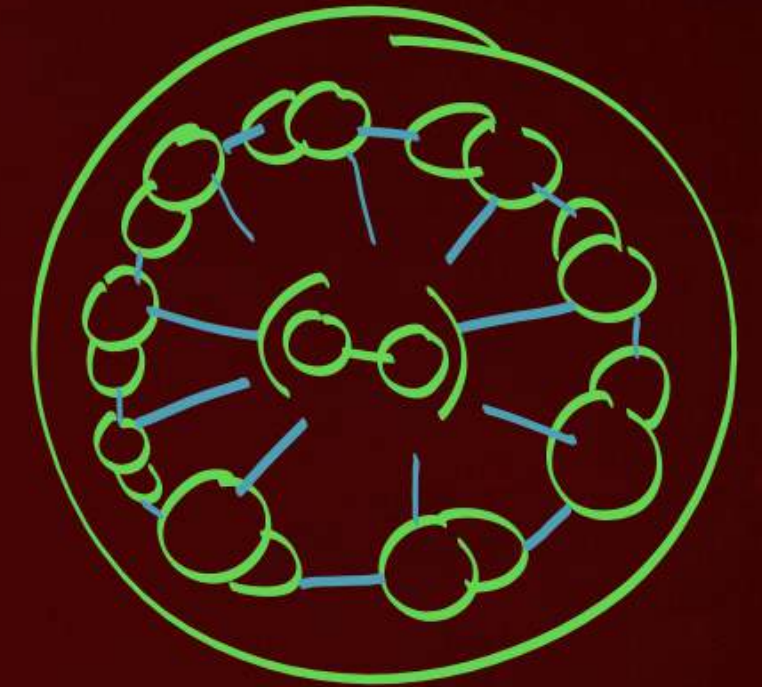
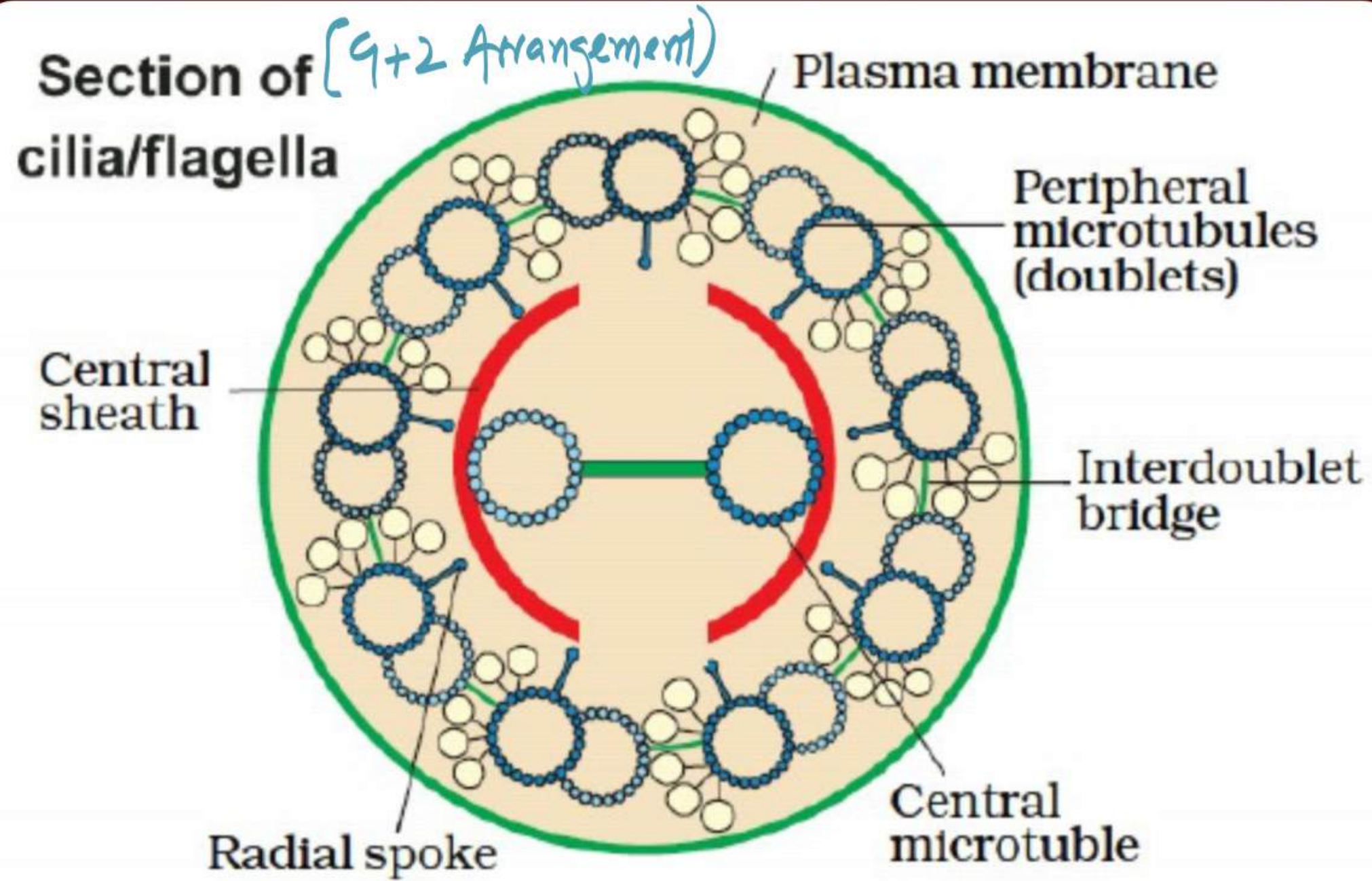


- ❑ It is a network of filamentous proteinaceous structures present in the cytoplasm.
- ❑ It provides mechanical support, motility, maintenance of the shape of the cell etc.

8.

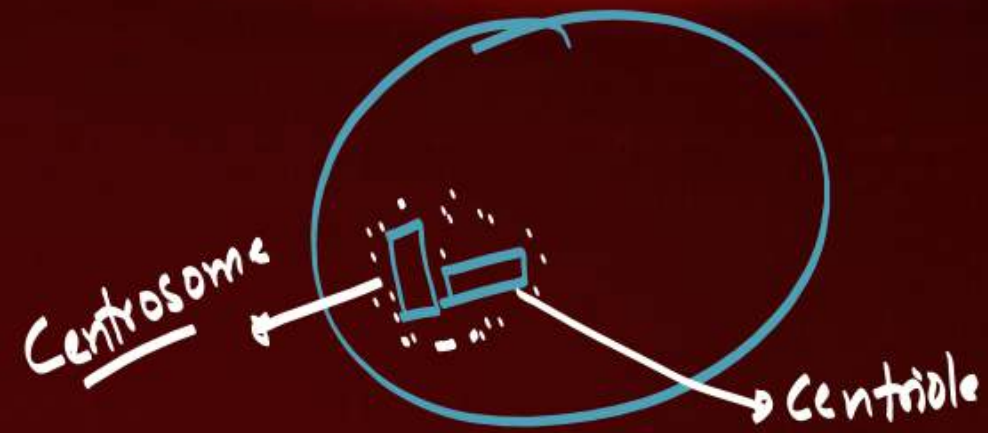
Cilia and Flagella

- ❑ They are hair-like outgrowths of the cell membrane.
- ❑ Cilia: Small structures which work like oars. Causes the movement of the cell or surrounding fluid.
- ❑ Flagella: Longer. Responsible for cell movement. Flagella of prokaryotes and eukaryotes are structurally different.





- ❑ Cilium and flagellum are covered with plasma membrane. Their core (**axoneme**) has many **microtubules** running parallel to the long axis.
- ❑ The axoneme has 9 pairs of doublets of radially arranged peripheral microtubules and a pair of central microtubules. This is called 9+2 array.
- ❑ The central tubules are connected by bridges and are enclosed by a central sheath. It is connected to one of the tubules of each peripheral doublet by a radial spoke. Thus, there are 9 radial spokes. The peripheral doublets are also interconnected by linkers. / Bridge
- ❑ Cilium and flagellum emerge from centriole-like structure called the basal bodies.



9. Centrosome and Centrioles

- ❑ Centrosome is an organelle usually containing two nonmembrane bound cylindrical structures called centrioles.
- ❑ They are surrounded by pericentriolar materials.
- ❑ The centrioles lie perpendicular to each other. They are made up of 9 evenly spaced peripheral fibrils of tubulin. Each of the peripheral fibril is a triplet. The adjacent triplets are also linked.

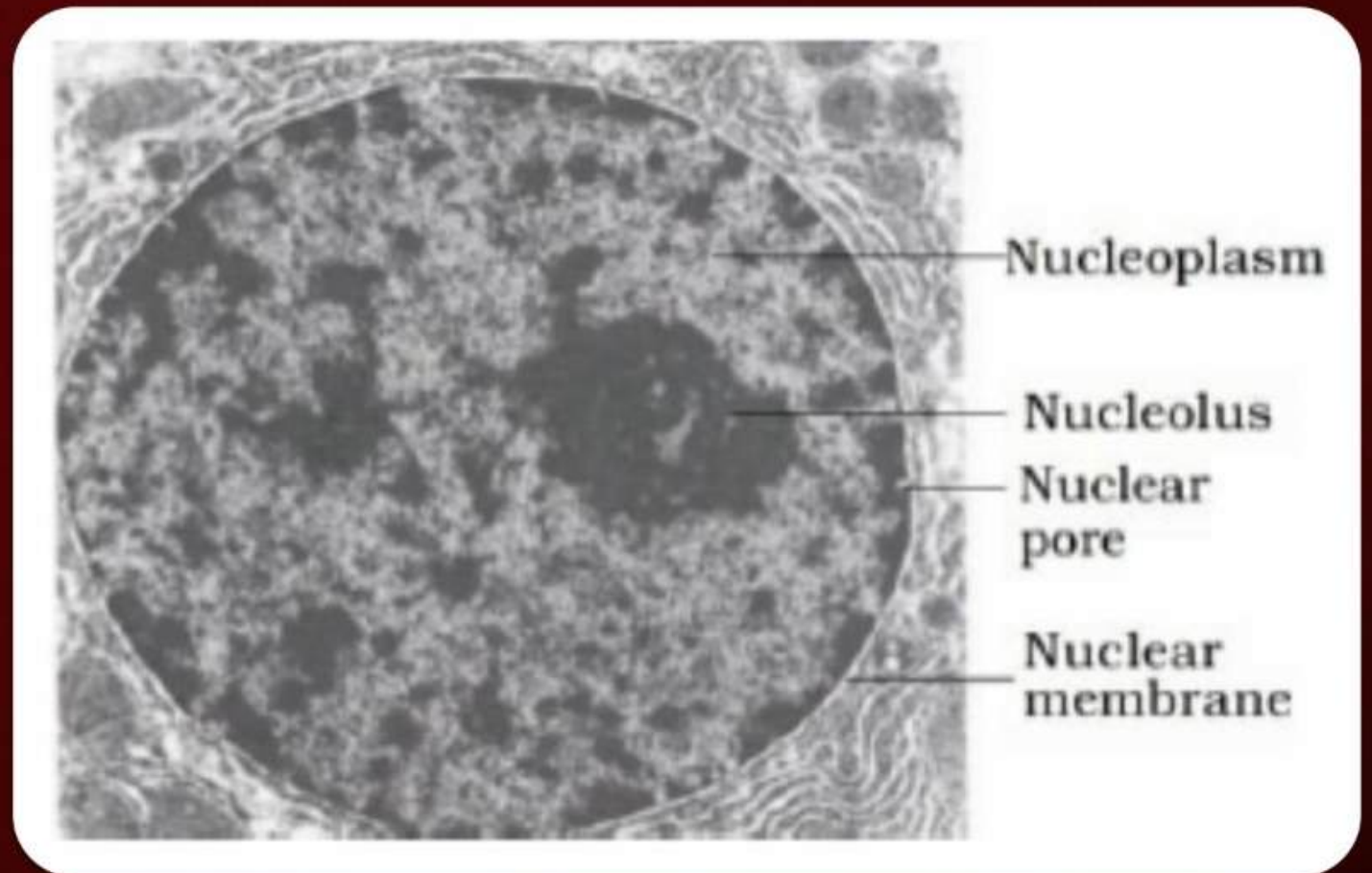


- ❑ The central part of the centriole is also proteinaceous and called the hub, which is connected with tubules of the peripheral triplets by radial spokes made of protein. *→ cilia flagella*
- ❑ The centrioles form the basal body of cilia or flagella, and spindle fibers that give rise to spindle apparatus during cell division in animal cells.



Nucleus

- ❑ Nucleus was first described by **Robert Brown (1831)**.
- ❑ The material of the nucleus stained by the basic dyes was given the name **chromatin** by **Flemming**.





□ Normally, a cell has only one nucleus. Some cells have more than one.
Some mature cells lack nucleus. E.g. mammalian RBC and sieve tube
cells of vascular plants.

□ The ^{rest}interphase nucleus contains

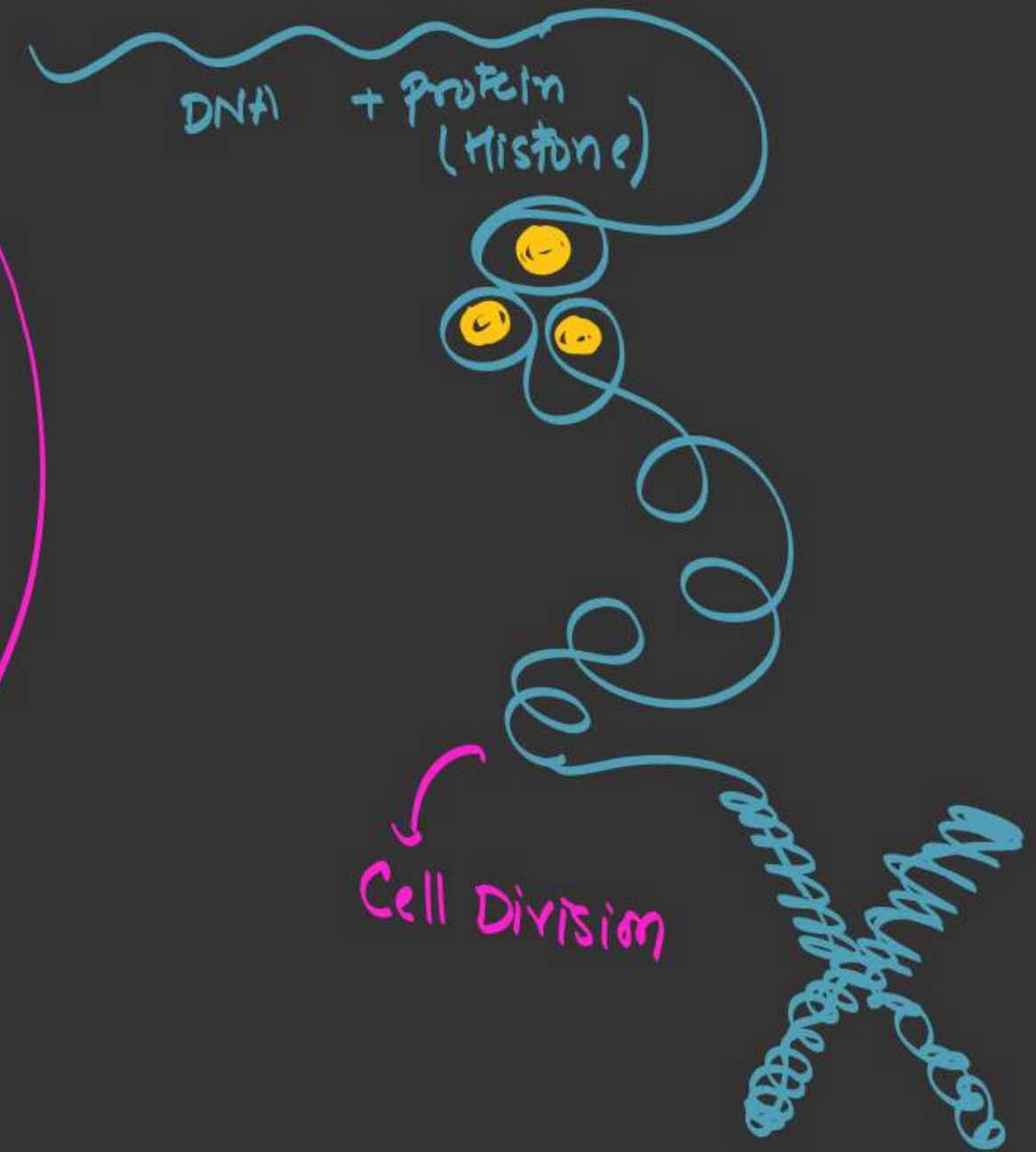
Nuclear envelope: Double layered membrane with a space between (10–50
nm) called perinuclear space. It is a barrier between the materials present in
nucleus & cytoplasm. Outer membrane usually remains continuous with ER
and also bears ribosomes on it.



- ❑ Nuclear envelope has minute pores formed by the fusion of its two membranes. These are the passages for the movement of RNA and protein between nucleus and cytoplasm.

Nuclear matrix (nucleoplasm)

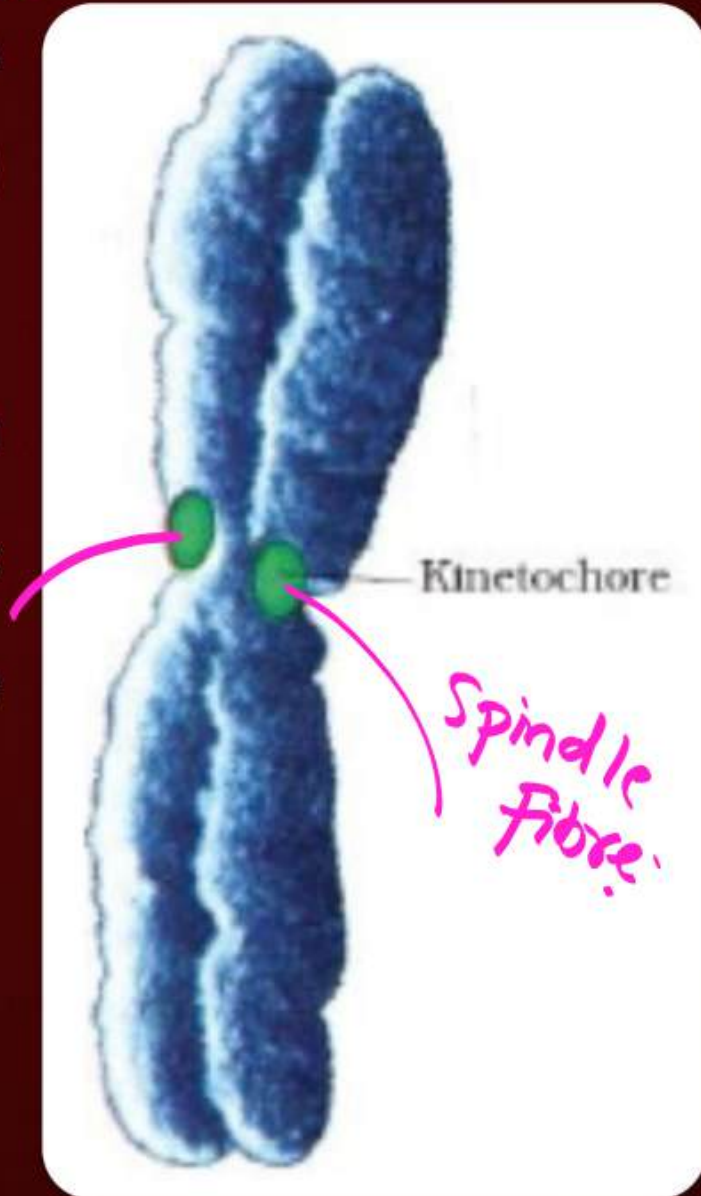
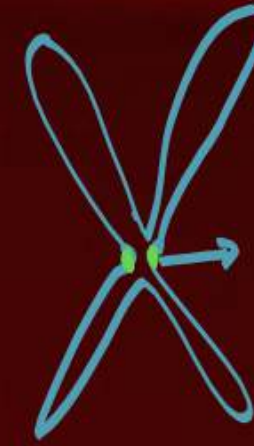
- ❑ Chromatin: A network of ^{DNA Protein} (nucleoprotein) fibres. It contains DNA and basic proteins (histones), non-histone proteins and RNA. During cell division, chromatins condense to form chromosomes.
- ❑ Nucleolus: One or more non-membranous spherical bodies. It is continuous with the nucleoplasm. It is a site for ribosomal RNA synthesis.

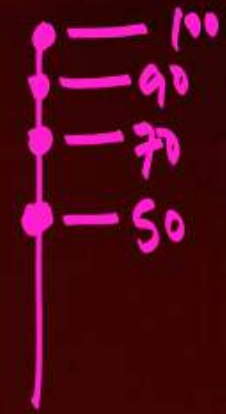




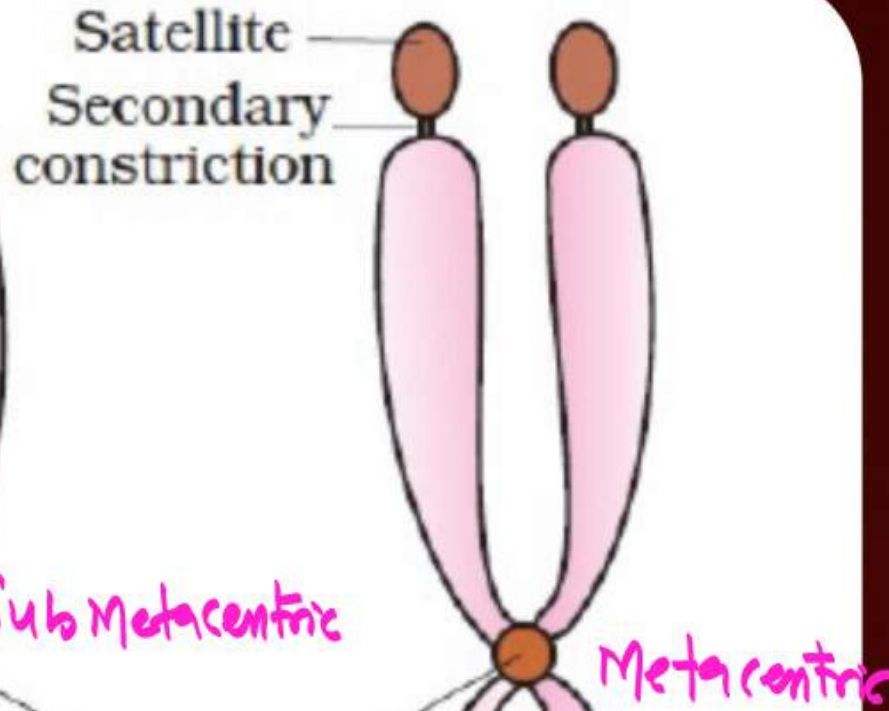
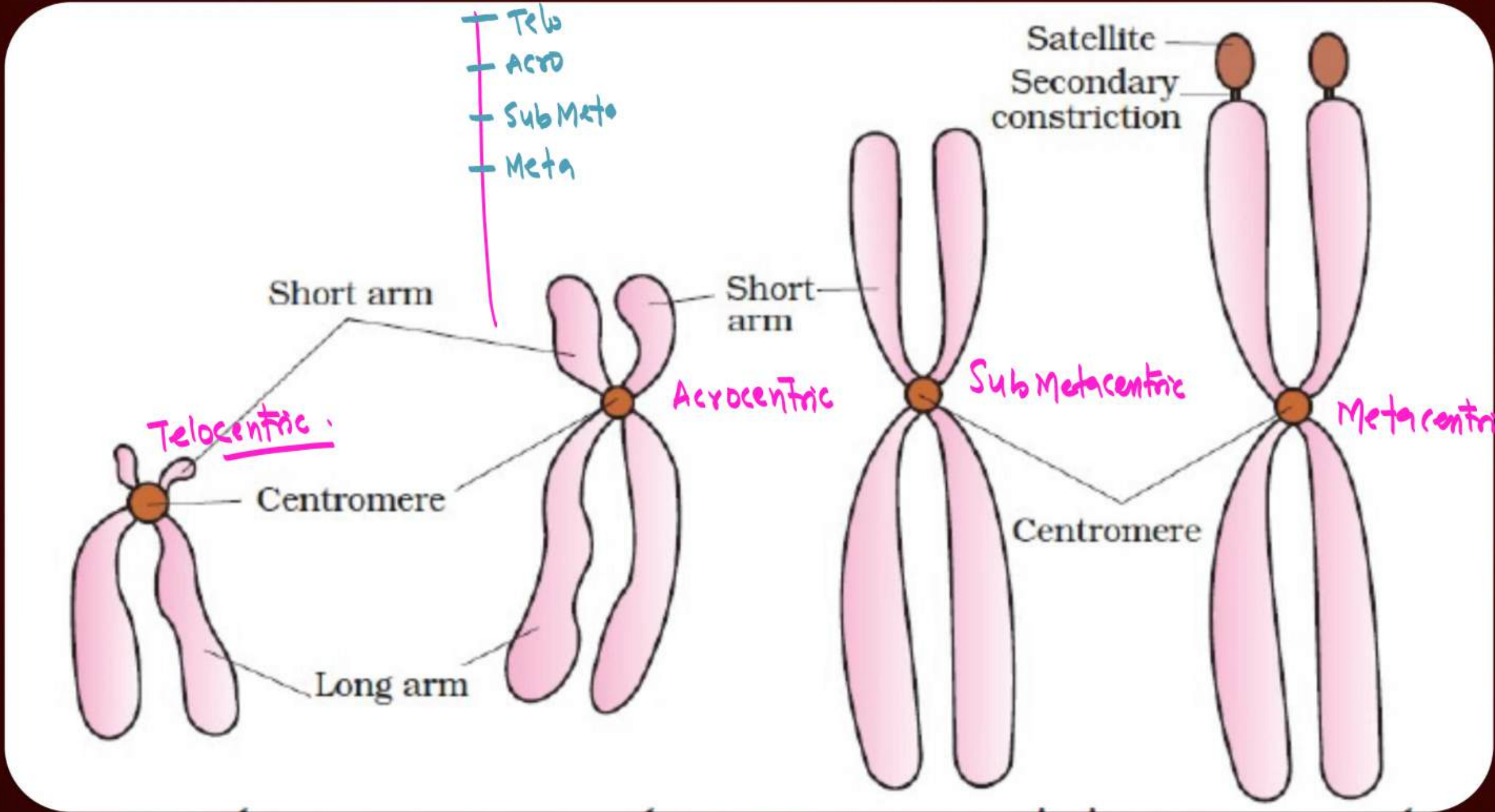
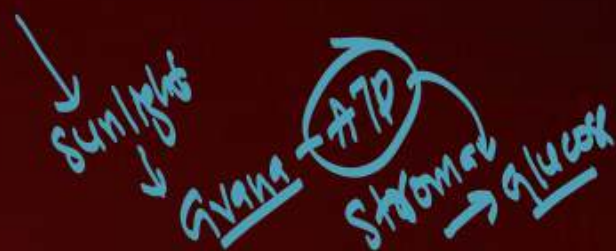
Chromosomes:

- A human cell has 2 m long thread of DNA distributed among its 46 (23 pairs) chromosomes.
- Every chromosome has a primary constriction (centromere). On the sides of centromere, disc shaped structures called kinetochores are present.





- ❑ Based on position of centromere, chromosomes are 4 types:
- ✓ **Metacentric chromosome:** Middle centromere forming two equal arms of the chromosome.
- ✓ **Sub-metacentric chromosome:** Centromere is nearer to one end forming one shorter arm and one longer arm.
- ✓ **Acrocentric chromosome:** Centromere is close to its end forming one very short and one very long arm.
- ✓ **Telocentric chromosome:** Terminal centromere.



Thank
You